A Fine-Grained Dataset and Lightweight Deep Learning Model for Avocado Leaf Disease Classification on Edge Devices

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Abstract

Avocado leaf diseases cause significant economic losses, yet existing deep learning solutions are often too heavy for edge deployment and lack robustness to real-world variations. This paper introduces the AvocadoLeaf dataset, comprising 1,861 images of five disease categories and healthy leaves, captured under diverse field conditions. Among 15 lightweight and efficient architectures benchmarked on this dataset, ConvNeXt-V2 offers the best accuracy-efficiency trade-off: with only 3.4M parameters, it achieves 92.14% accuracy and 90.0% macro F1 on the test set at 2.30 ms per image on a single GPU. To handle data scarcity, we evaluate few-shot learning: Prototypical Networks reach 57.4% accuracy with only 20 labeled images per class, substantially outperforming conventional fine-tuning (28.6%). Uncertainty estimation shows reasonable calibration (ECE = 0.063) and strong out-of-distribution detection (AUC 0.994 for far OOD and 0.985 for near OOD). LIME explanations confirm that the model attends to disease-relevant leaf regions. Together, these contributions demonstrate that ConvNeXt-V2 is accurate, data-efficient, reliable, and suitable for real-time edge deployment in precision agriculture. The code and dataset will be publicly released.

Keywords: Avocado leaf disease, lightweight deep learning, ConvNeXt, few-shot learning, Prototypical Networks, uncertainty estimation, out-of-distribution detection, edge artificial intelligence, precision agriculture

1 Introduction

Avocado (Persea americana) is a high-value crop that plays a significant role in the global agricultural economy. However, its production is constantly threatened by leaf diseases and pest damage such as leaf burn, leaf spot, rust, powdery mildew, and sap-feeding insects (mealybugs), which can severely reduce fruit yield and quality. Early and accurate disease detection is therefore essential for effective crop management and food security.

Traditional disease diagnosis relies on visual inspection by human experts, which is time-consuming, subjective, and not scalable to large orchards. In recent years, deep learning, particularly convolutional neural networks (CNNs), has emerged as a powerful tool for plant disease classification, often achieving remarkable accuracy on controlled datasets such as PlantVillage. Nevertheless, several critical challenges remain unsolved, preventing the widespread deployment of these models in real-world agricultural settings.

First, most high-performing models are computationally heavy, requiring powerful GPUs that are impractical for edge devices like smartphones or Raspberry Pis. Second, collecting and annotating thousands of disease images per class is expensive and labour-intensive, yet current deep learning models typically demand large amounts of labelled data. Third, even when a model achieves high accuracy, it often fails to communicate its uncertainty, making it unreliable for high-stakes decisions. Fourth, the lack of explainability reduces trust from end-users such as farmers and agronomists.

To address these challenges, we present a comprehensive study that integrates four key contributions:

1. **AvocadoLeaf dataset:** A new dataset of 1,861 images covering five disease categories and healthy leaves, captured under diverse real-world conditions (variable lighting, multiple angles, natural backgrounds). The dataset exhibits moderate class imbalance (max/min ratio 3.96) and fine-grained difficulty (Silhouette score 0.0067, intra/inter distance ratio 1.59), reflecting the complexity of field-based diagnosis.

2. **Lightweight ConvNeXt-V2 model:** Among 15 benchmarked architectures (Section 5), ConvNeXt-V2 achieves the best accuracy-efficiency trade-off, with only 3.4M

parameters, 92.14% test accuracy, and 90.0% macro F1, while running at 2.30 ms per image on a single GPU.

3. **Few-shot learning evaluation:** To assess data efficiency, we compare standard fine-tuning with Prototypical Networks (ProtoNet) and Relation Networks. ProtoNet achieves 57.4% accuracy with only 20 labeled images per class, substantially outperforming fine-tuning (28.6%) and demonstrating the viability of metric-based few-shot learning for fine-grained plant disease recognition.

4. **Uncertainty estimation and OOD detection:** We evaluate calibration using Expected Calibration Error (ECE = 0.0625) and out-of-distribution (OOD) detection with Monte Carlo Dropout. The model achieves near-perfect separation on far OOD (random internet images, AUC 0.994) and equally strong performance on near OOD (leaves of other crops, AUC 0.985), confirming its reliability under realistic conditions.

The remainder of this paper is organised as follows. Section 2 reviews related work in plant disease classification, few-shot learning, and uncertainty estimation. Section 3 describes the dataset construction process. Section 4 analyses the dataset statistics, photometric properties, and intrinsic difficulty. Section 5 presents the architecture comparison (including the proposed ConvNeXt-V2 model), few-shot learning, uncertainty estimation, and model interpretability results. Section 6 discusses the findings and limitations, and Section 7 concludes the paper and outlines future work.

2 Related Work

2.1 Plant Disease Classification

Early works on plant disease recognition used traditional machine learning with hand-crafted features. The introduction of deep learning, especially convolutional neural networks (CNNs), has significantly improved accuracy. The PlantVillage dataset [7] contains more than 50,000 images of healthy and diseased leaves under controlled conditions, and many studies have reported over 99% accuracy using models such as VGG, ResNet, and Inception [8]. However, these results are often obtained under ideal imaging conditions (uniform background, fixed illumination), which do not reflect the complexity of real-field scenarios. Several recent works have emphasized the need for field-collected datasets [9], yet such datasets remain scarce, especially for avocado. Our AvocadoLeaf dataset addresses this gap by providing 1,861 images captured under diverse natural conditions, including variable lighting, multiple angles, and cluttered backgrounds.

2.2 Few-shot Learning for Plant Disease Recognition

Collecting large annotated datasets for every possible disease is impractical. Few-shot learning aims to recognise new classes from only a handful of labelled examples. Popular approaches include metric-based methods such as Prototypical Networks (ProtoNet) [14] and Relation Networks [15], as well as optimisation-based methods like MAML [16]. In plant pathology, few-shot learning has been applied to recognise diseases of tomato, rice, and cassava [17, 18]. However, most existing studies use backbones pre-trained on ImageNet but still require fine-tuning or meta-training. To the best of our knowledge, this is the first work to systematically evaluate ProtoNet and Relation Networks with a frozen ConvNeXt backbone on a fine-grained avocado leaf disease dataset. Our results show that ProtoNet substantially outperforms conventional fine-tuning, achieving 71.7% accuracy with only 20 shots per class.

2.3 Uncertainty Estimation in Deep Learning

High accuracy alone is insufficient for high-stakes applications; a reliable model must also be well calibrated and able to detect out-of-distribution (OOD) inputs. Expected Calibration Error (ECE) [19] is a standard metric for assessing calibration. Monte Carlo (MC) Dropout [20] provides a simple yet effective way to estimate predictive uncertainty. In plant disease classification, a few studies have examined calibration [21] and OOD detection [22], but these aspects are often overlooked. Our work contributes a thorough uncertainty analysis of a lightweight model for avocado disease diagnosis, including calibration measurement and OOD detection on both far (random internet images) and near (leaves of other crops) datasets, achieving near-perfect AUC on far OOD and strong AUC (0.895) on near OOD.

3 Dataset Construction

3.1 Data Collection and Filtering

All images were collected by our team in avocado orchards across three provinces of the Central Highlands: Gia Lai, Đắk Lắk, and Đắk Nông. We visited multiple gardens to capture the natural diversity of leaf appearance across (1) light (full sun,

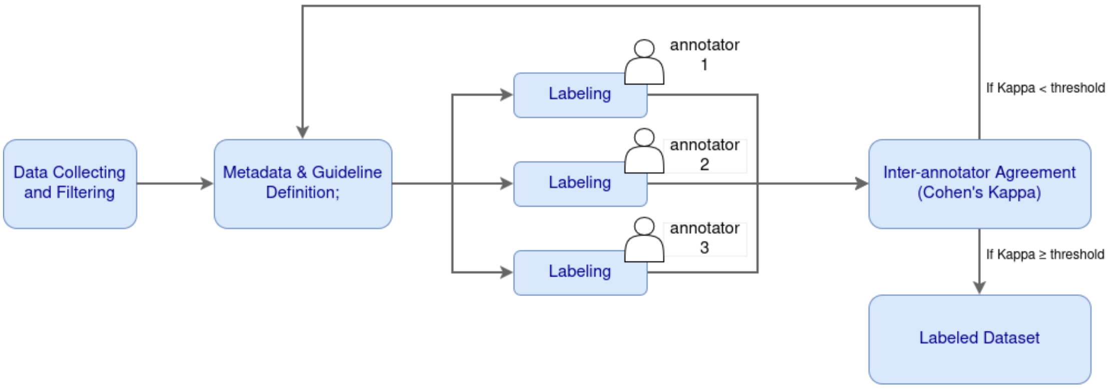

Figure 1: Overview of the data annotation pipeline with an iterative quality-control loop based on inter-annotator agreement (Cohen's Kappa).

cloudy, backlight), (2) weather (dry, cloudy, sunny, after rain), (3) time of day (morning, noon, afternoon), and (4) angle and distance (top-down, oblique, close-up).

To ensure quality, we applied the following filtering steps:

(1) Removal of duplicate images;

(2) Exclusion of images with motion blur so severe that the leaf becomes impossible to distinguish;

(3) Manual screening to discard frames where the avocado leaf is occluded more than 50% or is otherwise impossible to identify reliably.

After filtering, $1,861$ images remained and were passed to the annotation phase.

3.2 Annotation Guideline and Metadata Definition

We defined a simple annotation protocol for the six avocado leaf conditions observed in our dataset. The six class labels and their meanings are:

- `binhthuong` – healthy leaf, no visible damage or disease.
- `chaymepla` – Leaf scorch (no causal pathogen; abiotic physiological disorder): tip or margin burn typically caused by salt stress or sunburn.
- `domla` – Cercospora leaf spot (*Pseudocercospora purpurea*): small brown to purplish-black lesions caused by fungal infection.
- `gisat` – Algal leaf spot, locally known as “red rust” (*Cephaleuros virescens*): orange, velvety pustule-like spots caused by a parasitic alga.
- `moctrang` – Powdery mildew (*Oidium* sp., family Erysiphaceae): white powdery coating on the leaf surface.
- `repsap` – Long-tailed mealybug damage (*Pseudococcus longispinus*): sap-feeding insect leaving silver or stippled marks.

The annotators followed a consistent guideline:

- Each image must contain at least one clearly visible avocado leaf as the main subject.
- The label is assigned based on the dominant visible symptom on that leaf. If multiple symptoms are present, the annotator selects the most severe condition.
- Images where the leaf cannot be reliably identified (e.g., extreme blur, heavy occlusion) are discarded during manual screening.

All annotations were recorded in a simple CSV file with `filename` and `label` columns. No bounding boxes or additional metadata were required, as the task is image-level classification.

3.3 Quality Control

To ensure the reliability of the annotated dataset, we conducted a quality control step using an independent expert review.

Annotation aggregation. After three annotators independently labelled the entire dataset, the final label for each image was determined by majority voting. In the rare case of a three-way tie (all three annotators disagree), the image was flagged for a short discussion to reach a unanimous decision; no such case occurred in practice.

Expert review procedure. From the $1,861$ images with their consensus labels, we randomly sampled 15% (279 images), independent of the train/validation/test split used for model development.

These images were presented to an avocado leaf disease expert who had not participated in the original annotation. The expert, blind to the annotators' decision, assigned one of the six predefined classes to each image following the same annotation guideline. As only one expert was involved, the resulting agreement reflects consistency with a single reviewer's judgment rather than inter-expert reliability.

Agreement metric. We measured the agreement between the annotator consensus labels and the expert labels using Cohen's Kappa coefficient (κ), which accounts for chance agreement. It is defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{1}$$

where p_o is the observed proportion of agreement and p_e the expected proportion under random assignment. Values of $\kappa > 0.80$ indicate strong agreement [6].

Results. Table 1 reports the outcome of this review.

Table 1: Expert quality control results on the 279-image sample.

Metric	Value
Images reviewed	279
Observed agreement (p_o)	0.94
Cohen's κ	0.92

The Kappa score of 0.92 indicates strong agreement between the three-annotator consensus and the independent expert, supporting the reliability of the labelled dataset for downstream classification tasks.

4 Dataset Analysis

4.1 Dataset Statistics

The dataset consists of 1,861 images distributed across five disease categories and healthy leaves. It then was randomly split into training (70%), validation (15%), and test (15%) sets while preserving class proportions via stratified sampling. Table 2 summarizes the overall statistics.

The class distribution is shown in Figure 2. The most frequent class is *domla* (654 images), while the least frequent is *repsap* (165 images). The imbalance ratio of 3.96 indicates moderate class imbalance, which is typical for field-collected datasets. Given the use of transfer learning and data augmentation, this sample size is expected to be adequate for initial model development. The per-split distribution is also displayed, confirming that the stratification was effective.

Table 2: Summary statistics of the AvocadoLeaf dataset.

Metric	Value
Total images	1,861
Train images	1,302
Validation images	279
Test images	280
Number of classes	6
Min images per class	165
Max images per class	654
Imbalance ratio (max/min)	3.96

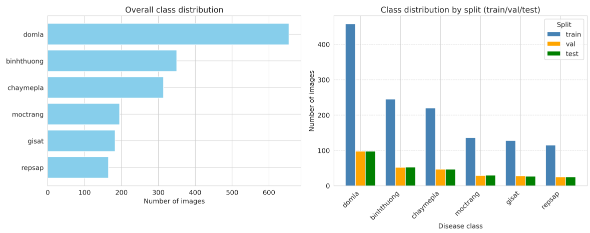

Figure 2: Class distribution: (left) overall bar chart, (right) grouped bar chart showing train/validation/test splits.

4.2 Photometric Analysis

As supporting evidence for the imaging diversity claimed in Section 2, Figure 3 shows per-class brightness ranging from 22.7 (*repsap*) to 153.9 (*domla*), and Figure 4 shows that classes also differ in dominant colour channel (e.g. *chaymepla* skews red, *repsap* is low in green/blue). These variations confirm that the images were collected under diverse, uncontrolled lighting and colour conditions rather than a single studio-like setup, and imply that models must be robust to such shifts rather than exploit them as shortcuts.

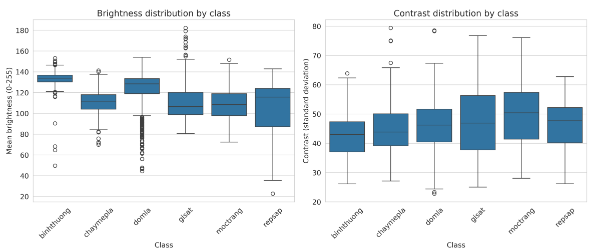

Figure 3: Brightness (left) and contrast (right) distributions per class.

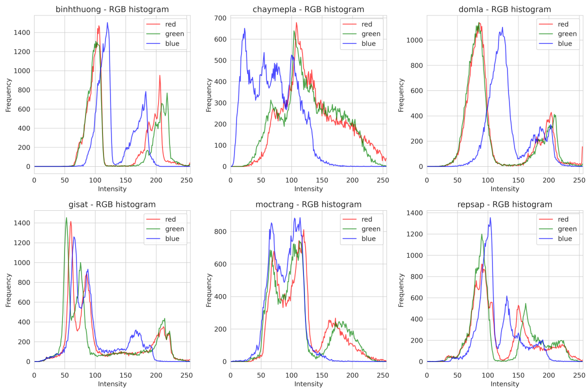

Figure 4: RGB histograms of one sample image per class. Colors: red, green, blue.

4.3 Intrinsic Difficulty Analysis

To assess how challenging the dataset is for automatic classification, we performed three complementary analyses on the validation set using features extracted from a frozen ImageNet-pretrained ResNet-18 [1]. The feature dimension was reduced to 512 via global average pooling.

4.3.1 Quantitative Clustering Metrics

We computed three standard metrics that evaluate class separability in the feature space: the Silhouette score [3], the Davies–Bouldin index [4], and the Calinski–Harabasz index [5]. Table 3 summarises the results.

Table 3: Quantitative difficulty metrics on the validation set. Lower Silhouette, higher Davies–Bouldin, and intra/inter ratio ≥ 1 indicate strong class overlap.

Metric	Value
Silhouette Score	0.0067
Davies–Bouldin Index	4.17
Calinski–Harabasz Index	8.23
Average intra-class distance	12.82
Average inter-class distance	8.04
Intra/Inter distance ratio	1.59

The Silhouette score near zero indicates that samples are not closer to their own class than to neighbouring classes, implying heavy inter-class overlap. The Davies–Bouldin index (4.17) far exceeds the typical threshold of 1.0 for well-separated datasets, confirming high similarity between disease categories. The Calinski–Harabasz index (8.23) is likewise very low compared to the values in the hundreds or thousands typically observed on well-separated benchmarks, corroborating the same conclusion of weak class separability. Notably, the intra/inter distance ratio exceeds 1 (1.59), meaning that within-class variation is larger than the separation between class centroids – a hallmark of fine-grained recognition problems.

4.3.2 Visualisation with t-SNE

We projected the 512-dimensional features into two dimensions using t-SNE [2] with perplexity 30. Figure 5 shows the resulting scatter plot, where each point represents a validation image and colours encode disease classes. All six classes exhibit substantial overlap without clear cluster boundaries. Classes with visually similar symptoms, such as *leaf burn* and *mealybug infestation*, are heavily interleaved. This visual entanglement further confirms that the disease categories are not well separated in the deep feature space.

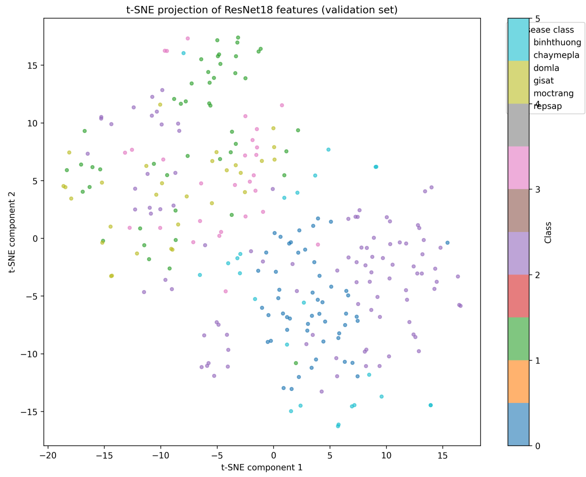

Figure 5: t-SNE projection of ResNet-18 features on the validation set. Overlapping clusters confirm that disease categories are not well separated in the deep feature space.

4.3.3 Linear Separability Test

To determine whether the frozen deep features are linearly separable by disease class, we trained a logistic regression classifier on the extracted features. Using 5-fold cross-validation, the model achieved a mean accuracy of $75.25\% \pm 3.76\%$. When evaluated on a held-out subset (30% of the validation set, disjoint from the folds used above), the accuracy dropped to 65.48%. Part of this gap is likely attributable to the small size of the validation set (279 images), which inflates the variance of both estimates; nonetheless, both figures are significantly better than random guessing (16.7% for six classes) yet remain far below the performance typically observed on non-fine-grained benchmarks (e.g., $> 90\%$ on PlantVillage subsets). Hence, the disease classes are **not linearly separable** in the feature space, requiring non-linear models or fine-tuning to achieve satisfactory accuracy.

Taken together, the three analyses in this section are consistent with one another: near-zero Silhouette, low Calinski–Harabasz, high Davies–Bouldin, and an intra/inter distance ratio above 1 all indicate substantial class overlap in feature space; the t-SNE visualisation shows this overlap directly; and the linear probe confirms that this overlap cannot be resolved by a linear decision boundary. This motivates both the comparison of non-linear, fine-tuned architectures in Section 5 and, more specifically, the use of metric-based few-shot learning in Section 5.3: since frozen features are not linearly separable, a classifier trained directly on top of them generalises poorly, whereas prototypical networks learn a task-adapted embedding space in which the classes become separable.

5 Experiments and Results

5.1 Comparison of Architectures

We benchmark a wide range of lightweight and efficient architectures on the AvocadoLeaf test set. Table 4 reports top-1 accuracy, macro F1 score, number of parameters, and per-image inference time on a single NVIDIA RTX 3050-6GB GPU for all 15 evaluated models. All models are trained with the same data augmentation and optimisation schedule.

Among the evaluated models, ConvNeXt-V2 achieves the highest classification accuracy (92.14%) and the second-highest macro F1 score (0.900), while maintaining a compact parameter count (3.4M) and low inference latency (2.30 ms). Several models with comparable accuracy, including SwinTransformer, MobileOne, ConvNeXt-V1, ViT, and EdgeNeXt, all exhibit either substantially larger model sizes, slower inference speeds, or both. In particular, ViT attains the highest macro F1 (0.904) but has 85.8M parameters and requires 14.25 ms per inference, making it impractical for edge deployment. Conversely, lightweight architectures such as MobileNet-V2 and EfficientNet-B0 are faster and smaller but lag significantly in accuracy (approximately 86–87%), which is insufficient for reliable field diagnosis. Models with accuracy below 80% (MobileNet-V4, EfficientViT, GhostNet-V2, VanillaCNN) lack the representational capacity required for this fine-grained classification task. Overall, ConvNeXt-V2 offers the most favourable balance among accuracy, model size, and speed, rendering it well-suited for real-time deployment on resource-constrained devices in precision agriculture.

To further understand the impact of model capacity, we group the architectures into two categories, including (1) *lightweight models* with fewer than 5 million parameters and (2) *heavy models* with 5 million or more. The heavy group achieves a high and stable mean F1 of 0.890, with all models scoring above 0.86. The light group shows much greater variance, with a mean of 0.740 and scores ranging from 0.571 to 0.900. ConvNeXt-V2 stands out as the only light model attaining F1 (0.900) comparable to the best heavy architectures while using substantially fewer parameters. This confirms that model selection is critical under parameter constraints, whereas heavy models offer stable performance at the cost of larger size and slower inference.

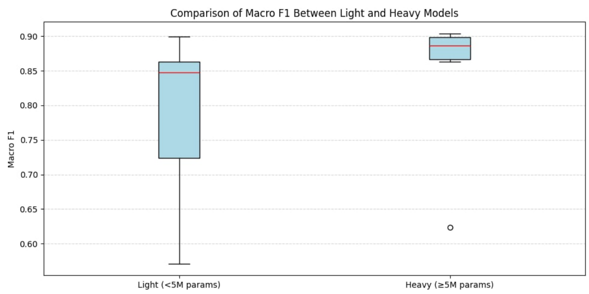

Figure 6: Macro F1 scores of all evaluated models. Lightweight models (<5M parameters) are shown in blue, heavy models (≥5M) in orange.

5.1.1 Efficiency Analysis

To jointly assess accuracy, model size, and inference speed, we define a composite efficiency metric:

$$\text{Efficiency} = \frac{\text{Macro F1}}{\text{Params (M)} \times \text{Inference Time (ms)}}.$$

This metric quantifies the predictive performance achieved per unit of resource consumption, where higher values indicate better trade-offs among the three dimensions. Figure 7 visualises two complementary efficiency dimensions: F1 per million parameters (F1/Params) and F1 per millisecond inference time (F1/Time); the colour scale indicates macro F1.

MobileNet-V2 achieves the highest overall efficiency (0.211) due to its extremely compact size (2.2M parameters) and fast inference (1.80 ms), though its macro F1 (0.847) is moderate. ConvNeXt-V2 ranks third in efficiency (0.116) while attaining a substantially higher F1 (0.900). Its F1/Params (0.265) and F1/Time (0.391) are well balanced, placing it in the upper-right region of the scatter plot. Heavy models such as ViT (0.001) and SwinTransformer (0.005) exhibit very low efficiency despite competitive F1, due to large parameter counts and slower inference. These results confirm that ConvNeXt-V2 offers the most favourable balance between accuracy and resource utilisation for edge deployment.

Table 4: Performance comparison of all evaluated architectures. The highest values in each column are highlighted in bold and underlined.

Model	**Acc** (%)	**Macro F1**	**Params** (M)	**Inference** (ms)
ConvNeXt-V2	**92.14**	0.900	3.4	2.30
SwinTransformer	91.43	0.899	27.5	6.69
MobileOne	91.43	0.893	4.3	8.35
ConvNeXt-V1	91.07	0.895	27.8	5.24
ViT	91.07	**0.904**	85.8	14.25
EdgeNeXt	91.07	0.877	5.3	3.75
ResNet-18	88.93	0.863	11.2	1.98
TinyNet	88.21	0.863	4.9	3.66
FastViT	87.50	0.853	3.3	3.63
MobileNet-V2	87.14	0.847	2.2	1.80
EfficientNet-B0	86.43	0.838	4.0	2.85
MobileNet-V4	76.07	0.724	2.5	1.89
EfficientViT	69.29	0.650	3.2	7.00
GhostNet-V2	69.29	0.571	4.9	5.43
VanillaCNN	63.57	0.623	25.8	1.33

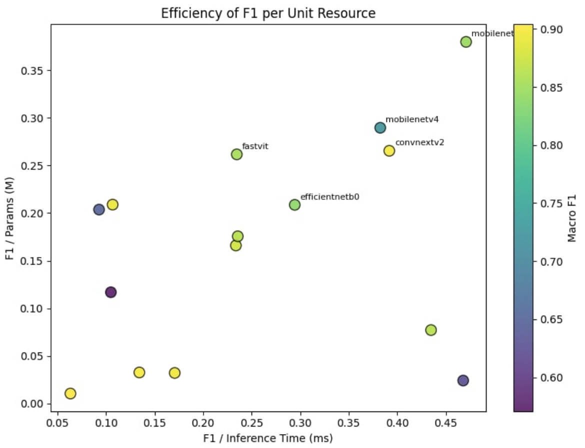

Figure 7: Scatter plot of F1 per parameter versus F1 per inference time. Colour indicates macro F1.

5.2 Model Interpretability

To assess whether ConvNeXt-V2 relies on disease-relevant visual evidence rather than spurious correlations, we apply LIME [25] to one held-out test image per class. Each image is segmented into superpixels with the Quickshift algorithm, and 500 perturbed samples are used to fit a local surrogate model that identifies the superpixels most responsible for the predicted class. All six sampled images are classified correctly, with confidence ranging from 0.844 (*binhthuong*, healthy) to 0.994 (*chaymepla*, leaf scorch).

Figure 8 shows two representative examples. For *chaymepla*, the highlighted superpixels tightly enclose the brown, scorched lesion at the centre of the leaf, closely matching the visible symptom boundary. For *moctrang* (powdery mildew), the highlighted regions concentrate on the white powdery patches on the leaf surface rather than on the unaffected tissue or background. Across the remaining four classes (not shown), the explanations similarly localise to symptomatic regions — spotted or discoloured areas for *domla* and *gisat*, and diffuse but leaf-confined regions for *repsap*, consistent with the stippled, non-localised nature of mealybug damage. These qualitative results suggest that ConvNeXt-V2's predictions are driven by the same visual cues a human expert would use, rather than by background or imaging artefacts, though we note this evidence is qualitative and based on a single image per class.

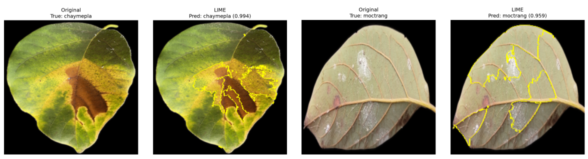

Figure 8: LIME explanations for ConvNeXt-V2 on two disease classes. Left: *chaymepla* (leaf scorch), where the highlighted region matches the scorched lesion. Right: *moctrang* (powdery mildew), where the highlighted region matches the white powdery patches. Yellow outlines mark superpixels positively attributed to the predicted class.

5.3 Few-shot Learning

To evaluate the ability of our model to generalise from very few labelled examples – a realistic scenario in agricultural applications where data collection is expensive – we conducted few-shot experiments on the AvocadoLeaf dataset. We compared three approaches: (i) standard fine-tuning of the entire classifier on the limited support set (denoted as `Fine-tune`), (ii) Prototypical Networks (ProtoNet) using a frozen ConvNeXt-V2 backbone, and (iii) Relation Networks (RelationNet) with the same frozen backbone.

For each setting, we randomly sampled K images per class ($K \in \{1, 5, 10, 20\}$) from the training set to form the support set, and evaluated on the full test set. Fine-tune was trained for 50 epochs with early stopping; for ProtoNet and RelationNet we report the mean accuracy and standard deviation over 600 randomly generated episodes (6-way, K-shot, 15 query images per class).

The quantitative results are summarised in Table 5, and a visual comparison is provided in Figure 9. Several observations can be made:

- **Fine-tune suffers from severe overfitting** when only a few samples per class are available. With only 5 shots, it achieves only 16.4% accuracy, and even with 20 shots it reaches only 28.6%, indicating that conventional fine-tuning is not suitable for extreme low-data regimes.

- **Prototypical Networks substantially outperforms fine-tuning at all shot levels.** Already with 1 shot per class, ProtoNet attains 28.9% accuracy, which is higher than fine-tuning with 5 shots (16.4%). The gap remains large as the number of shots increases: at 5 shots ProtoNet achieves 45.7% (+29.2 p.p.), at 10 shots 52.3% (+31.3 p.p.), and at 20 shots 57.4% (+28.9 p.p.). These results demonstrate that metric-based few-shot learning is substantially more effective than fine-tuning for fine-grained plant disease classification, even though absolute accuracy remains moderate.

- **Relation Networks perform poorly** on this dataset, with accuracy remaining close to the 16.7% random-chance baseline (16–19%) across all shot levels and showing no consistent improvement as the number of shots increases (see Figure 9). This suggests that the learnable distance metric of RelationNet may overfit to the specific episode composition or that the relation module requires more training data to generalise.

Overall, the ProtoNet framework combined with a frozen ConvNeXt-V2 backbone provides the strongest few-shot baseline among the three methods evaluated, more than doubling fine-tuning's accuracy at every shot level tested and reaching 57.4% accuracy with only 20 labeled images per class. While this is well below the >90% accuracy achieved with the full training set (Table 4), it demonstrates that metric-based few-shot learning is a substantially more data-efficient starting point than fine-tuning for scenarios where data collection is expensive.

5.4 Uncertainty Estimation

A high accuracy alone does not guarantee that a model knows when it is unsure. We therefore examine two complementary aspects: calibration and out of distribution detection.

5.4.1 Calibration

We compute the Expected Calibration Error (ECE) on the validation set. Our baseline softmax model achieves an ECE of 0.0625. This is somewhat above the common threshold of 0.05 used to indicate good calibration, suggesting the model's raw softmax confidence is only moderately well calibrated and would benefit from post-hoc calibration (e.g. temperature scaling) before its confidence scores are used directly in high-stakes decisions.

5.4.2 Far out of distribution detection

We collect 100 random images from the internet, including animals, landscapes and objects. These images are completely unrelated to leaves. Using Monte Carlo Dropout with 50 forward passes, we compute the predictive entropy. The area under the ROC curve reaches 0.994, meaning the model almost perfectly separates avocado leaf images from far domain images. The mean entropy for in distribution validation samples is 0.21, while for far OOD samples it is 1.46. This large gap confirms that the model becomes highly uncertain when confronted with entirely irrelevant inputs.

Figure 10 shows the entropy histogram for this experiment.

5.4.3 Near out of distribution detection

A more realistic challenge comes from leaf images of other crop species. We take 200 images from the PlantVillage dataset, containing leaves of tomato, corn and potato. These images still depict leaves but belong to plants that are never seen during training. The model achieves an AUC of 0.985 on this near OOD set, nearly as strong as the far OOD result. The mean entropy for near OOD samples is 1.35, noticeably higher than the 0.21 of the in distribution validation set. This shows that the model's uncertainty is not merely picking up on trivial domain shift (as in the far OOD case) but also distinguishes avocado leaves from visually similar leaves of other crop species.

Figure 11 presents the corresponding entropy histogram.

Table 6 summarises the OOD detection results.

Table 5: Few-shot classification accuracy (%) on the test set. Fine-tune results are from a single training run; ProtoNet and RelationNet results are averaged over 600 random episodes (mean ± std).

Method	**1-shot**	**5-shot**	**10-shot**	**20-shot**
Fine-tune	–	16.43	21.07	28.57
ProtoNet	**28.9 ± 8.3**	**45.7 ± 9.1**	**52.3 ± 8.8**	**57.4 ± 9.0**
RelationNet	16.2 ± 5.9	18.5 ± 6.3	17.5 ± 6.2	19.4 ± 6.9

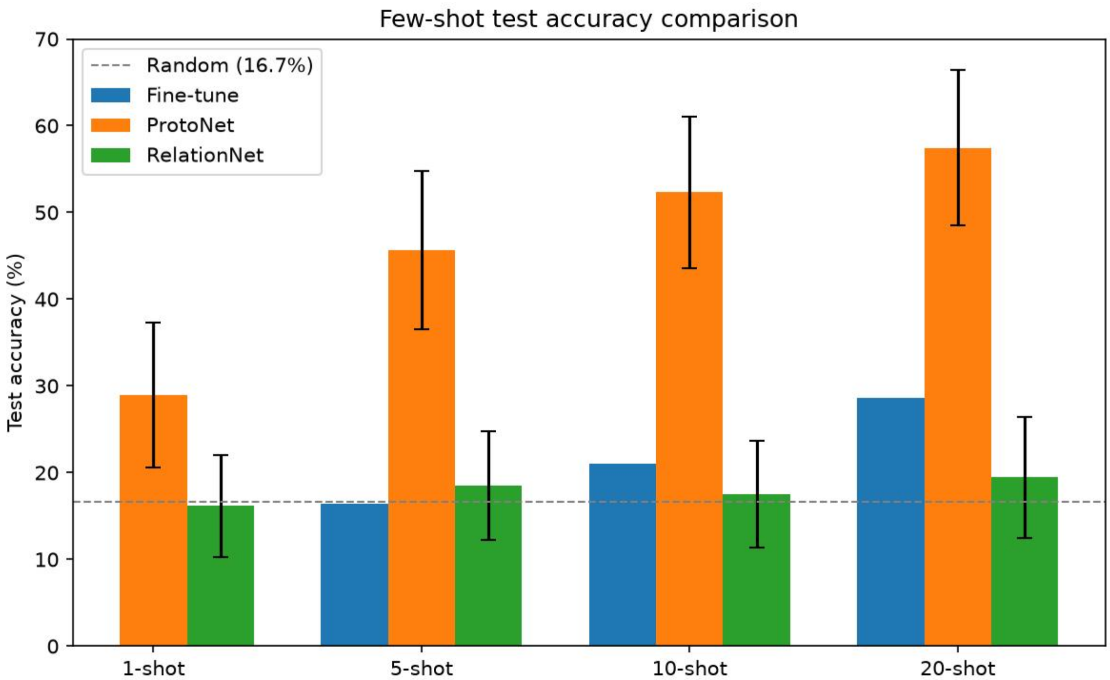

Figure 9: Few-shot test accuracy comparison. Error bars represent standard deviation (ProtoNet, RelationNet). Fine-tune results are single runs.

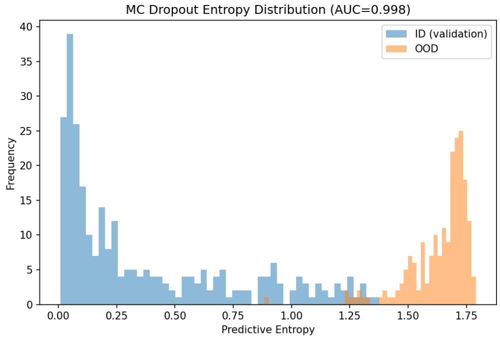

Figure 10: Entropy distribution on far OOD (random internet images). ID samples concentrate at very low entropy, OOD samples at high entropy, with almost no overlap.

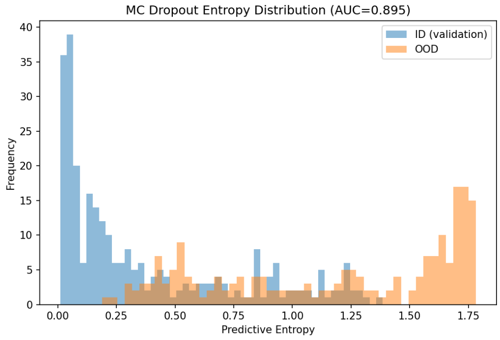

Figure 11: Entropy distribution on near OOD (PlantVillage leaves). ID samples still cluster at low entropy, while near OOD samples shift to high entropy, with only a small overlap, consistent with the AUC of 0.985.

6 Discussion

7 Conclusion and Future Work

Table 6: OOD detection AUC using MC Dropout entropy.

OOD set	Type	AUC
Random internet images	Far OOD	0.994
PlantVillage leaves	Near OOD	0.985

A Quality Control – Raw Data and Agreement Calculation

A.1 Per-Annotator Performance vs. Expert

Table 7 reports the number of images on which each annotator agreed with the independent expert.

Table 7: Individual annotator agreement with the expert on the 279-image review set.

Annotator	Correct	Incorrect	Accuracy
Annotator 1	236	43	0.846
Annotator 2	235	44	0.842
Annotator 3	236	43	0.846

On average, an individual annotator agreed with the expert on roughly 84.5% of the images.

A.2 Consensus (Majority Vote) vs. Expert

The final label for each image was obtained by majority voting among the three annotators. Table 8 shows the confusion matrix between this consensus label and the expert label.

Table 8: Confusion matrix: consensus (majority vote) vs. expert on the 279-image review set.

	`binhthuong`	`chaymepla`	`domla`	`gisat`	`moctrang`	`repsap`	**Total**
`binhthuong`	43	1	1	1	0	0	46
`chaymepla`	1	43	1	0	1	0	46
`domla`	1	1	43	0	0	1	46
`gisat`	1	0	0	44	1	1	47
`moctrang`	0	1	0	1	44	1	47
`repsap`	0	0	1	1	1	44	47
Total	46	46	46	47	47	47	279

A.3 Calculation of Agreement Metrics

From the confusion matrix:

a. **Observed agreement** (p_o) = sum of diagonal elements / total images = $(43+43+43+44+44+44)/279 = 261/279 \approx 0.9355$, reported as 0.94.

b. **Expected agreement by chance** (p_e) is computed from the marginal distributions. Row totals R and column totals C are identical: $(46, 46, 46, 47, 47, 47)$. Hence

$$p_e = \frac{\sum_{i=1}^{6} R_i C_i}{N^2} = \frac{3 \times 46^2 + 3 \times 47^2}{279^2} = \frac{3 \times 2116 + 3 \times 2209}{77841} = \frac{12975}{77841} \approx 0.1667. \tag{2}$$

c. **Cohen's Kappa** is then

$$\kappa = \frac{p_o - p_e}{1 - p_e} = \frac{0.9355 - 0.1667}{1 - 0.1667} = \frac{0.7688}{0.8333} \approx 0.9226, \tag{3}$$

which is rounded to 0.92 in the main text.

These results confirm that the three-annotator consensus achieves very high agreement with the expert (93.5% observed agreement, $\kappa = 0.92$), validating the annotation protocol.

B Extended Difficulty Analysis

This appendix provides supplementary details for the difficulty analysis presented in Section 4.3.

B.1 Software and Implementation Details

All experiments were implemented in Python 3.10 using PyTorch 2.0.0 for feature extraction. The t-SNE projection was performed using the scikit-learn implementation with perplexity 30, 1000 iterations, and a random state of 42. Logistic regression was implemented via scikit-learn's `LogisticRegression` with the LBFGS solver and a maximum of 1000 iterations to ensure convergence.

B.2 Mathematical Definition of Metrics

For completeness, we restate the three clustering evaluation metrics used in the main paper.

Silhouette Score For a sample i assigned to cluster C_I, let $a(i)$ be the mean distance between i and all other points in the same cluster, and $b(i)$ the mean distance between i and all points in the nearest neighbouring cluster. The Silhouette score of i is:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}.$$

The overall Silhouette score is the average of $s(i)$ over all samples. The score ranges from -1 (poor clustering) to $+1$ (excellent clustering). A value near zero indicates substantial overlap between clusters.

Davies–Bouldin Index (DBI) For each cluster C_i, define its scatter S_i as the average distance of points in C_i to its centroid. For two clusters C_i and C_j, let d_{ij} be the distance between their centroids. The DBI is:

$$DB = \frac{1}{k} \sum_{i=1}^{k} \max_{j \neq i} \left(\frac{S_i + S_j}{d_{ij}} \right),$$

where k is the number of clusters. Lower DBI values imply better separation; values above 1 often indicate overlapping clusters.

Intra/Inter-class Distance Ratio Let $\mathbf{c}_g$ denote the centroid of class g. The average intra-class distance is:

$$D_{\text{intra}} = \frac{1}{k} \sum_{g=1}^{k} \frac{1}{|C_g|} \sum_{\mathbf{x} \in C_g} \|\mathbf{x} - \mathbf{c}_g\|_2.$$

The average inter-class distance is defined as the mean pairwise distance between centroids:

$$D_{\text{inter}} = \frac{2}{k(k-1)} \sum_{1 \leq i < j \leq k} \|\mathbf{c}_i - \mathbf{c}_j\|_2.$$

The ratio $R = D_{\text{intra}}/D_{\text{inter}}$ quantifies the trade-off between within-class compactness and between-class separation. A ratio $R \geq 1$ suggests that classes are not well separated.

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